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In [8]: """
Demonstration of my UniversalClassifier wrapper, based on Scikit-Learn and
supporting GPU
"""

import warnings
import pandas as pd
from matplotlib import pyplot as plt
from ac.ml.classification import UniversalClassifier
from ac.tabular import Report

warnings.filterwarnings("ignore")
```

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In [9]: """
Loading a scraped dataset containing data about the top 500 billionaires
"""

billionaires = pd.read_csv("./billionaires.csv")
billionaires.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 499 entries, 0 to 498
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Rank                   499 non-null   int64
1   Name                   499 non-null   object
2   Wealth                 499 non-null   float64
3   Country                499 non-null   object
4   Sector                 499 non-null   object
5   Region                 499 non-null   object
6   Mapped_Region          499 non-null   float64
7   Mapped_Country         499 non-null   float64
8   Mapped_Sector          499 non-null   float64
dtypes: float64(4), int64(1), object(4)
memory usage: 35.2+ KB
```

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In [10]: """
Let's define a "sample", the 501st richest man in the world, and let's see
predicts his country the best
"""

algorithms = ["decisiontree", "randomforest", "svm"]
columns = ["Wealth", "Mapped_Region", "Mapped_Country", "Mapped_Sector"]
sample = [80, 1.0, 3.0]
```

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In [11]: """
UniversalClassifier explores automatically the optimal hyperparameter combination
Let's define the hyperparameter range
"""

grids = {
    "decisiontree": {
        "max_depth": [1, 2, 4, 8, 16, 32],
        "min_samples_leaf": [1, 2, 3, 4, 5],
        "criterion": ["gini", "entropy", "log_loss"],
    },
    "randomforest": {
        "max_depth": [1, 2, 4, 8, 16, 32],
        "min_samples_leaf": [1, 2, 3, 4, 5],
        "criterion": ["gini", "entropy", "log_loss"],
    }
}
```

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}
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In [12]: """
Launching the classification task. This cell demonstrates also my Report
which allows the generation of simple reports from tabular data in differ
formats.
"""
for algorithm in algorithms:
    uc = UniversalClassifier(method=algorithm, data=billionaires, subset=
uc.split(ratio="30%"))

    # Exploring different configurations with cross-validation
    param_grid = grids.get(algorithm, None)
    if param_grid is not None:
        uc.explore(param=param_grid, cv=5)

    # Fitting and performing cross-validation
    uc.fit(cv=5)
    predicted = uc.predict(sample)

    # Plotting the algorithm performance
    if param_grid is not None:
        uc.draw("performance", figsize=(10, 8), rotation=90)
        plt.show()

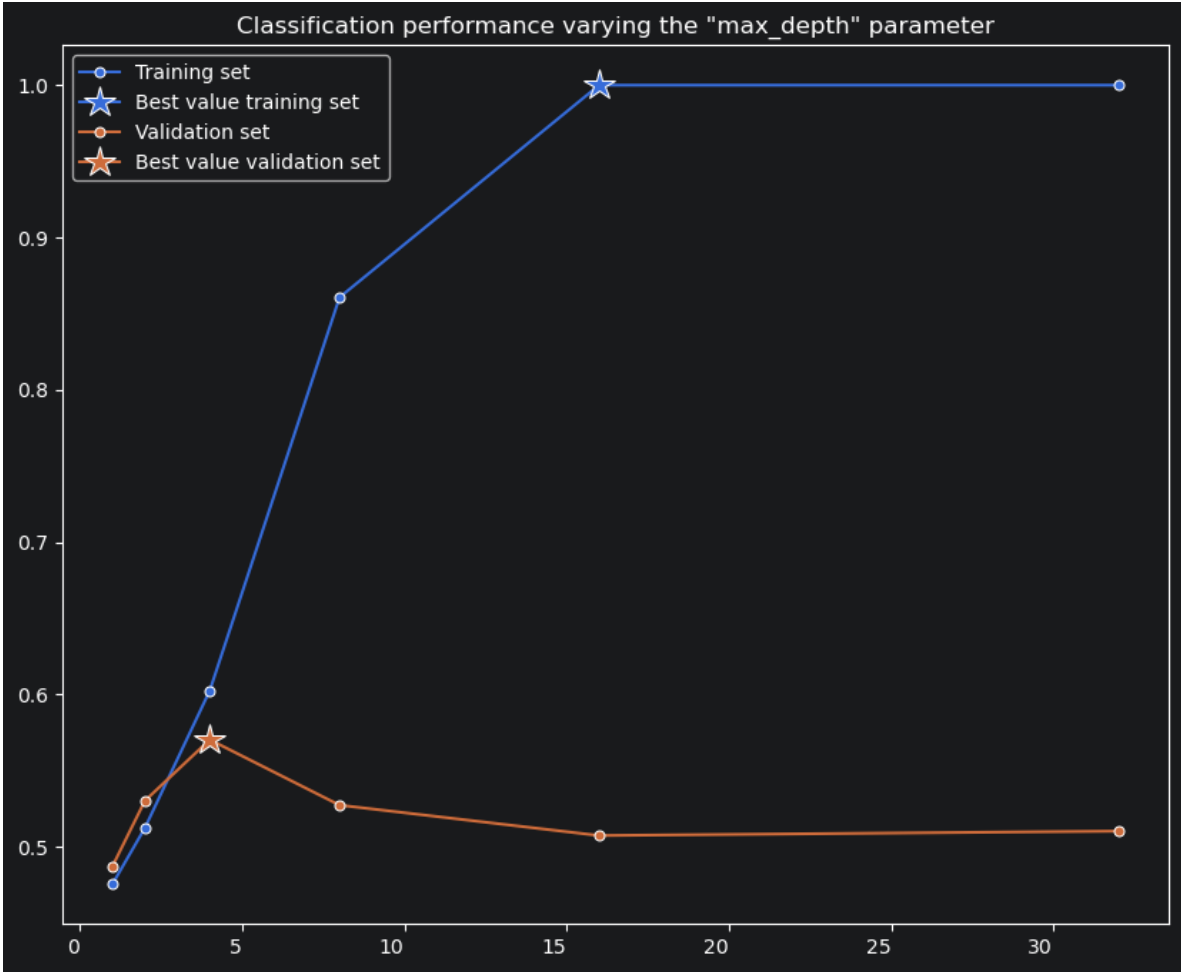
    # Generating some textual reports
    results = uc.evaluate("scores")
    title = "Scores of " + algorithm
    report = Report()
    report.set(source=results, head="Scores")
    report.headline(title)
    report.textualize()

    if algorithm != "svm":
        results = uc.evaluate("params")
        title = "Parameter values of " + algorithm
        report.set(source=results, head="Parameters")
        report.headline(title)
        report.textualize()

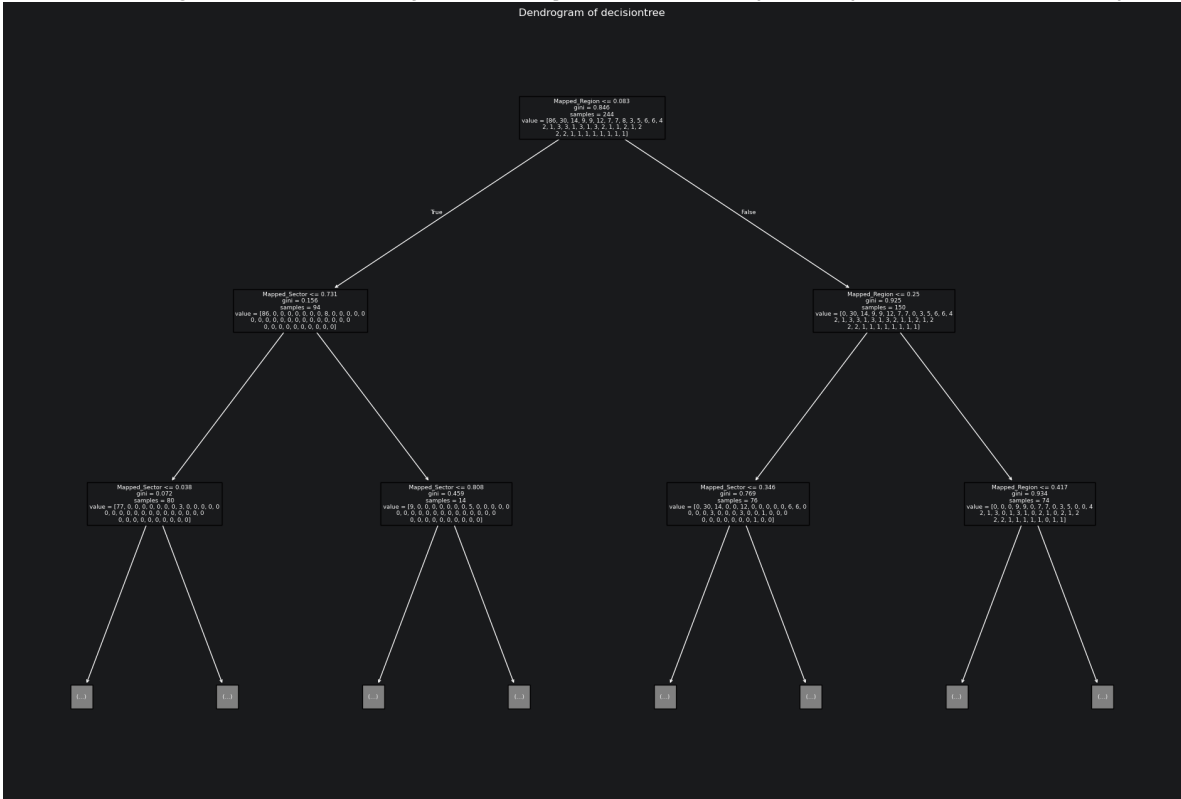
    # Saving everything in pdf
    report.dump(report_name=algorithm, extension=".pdf")

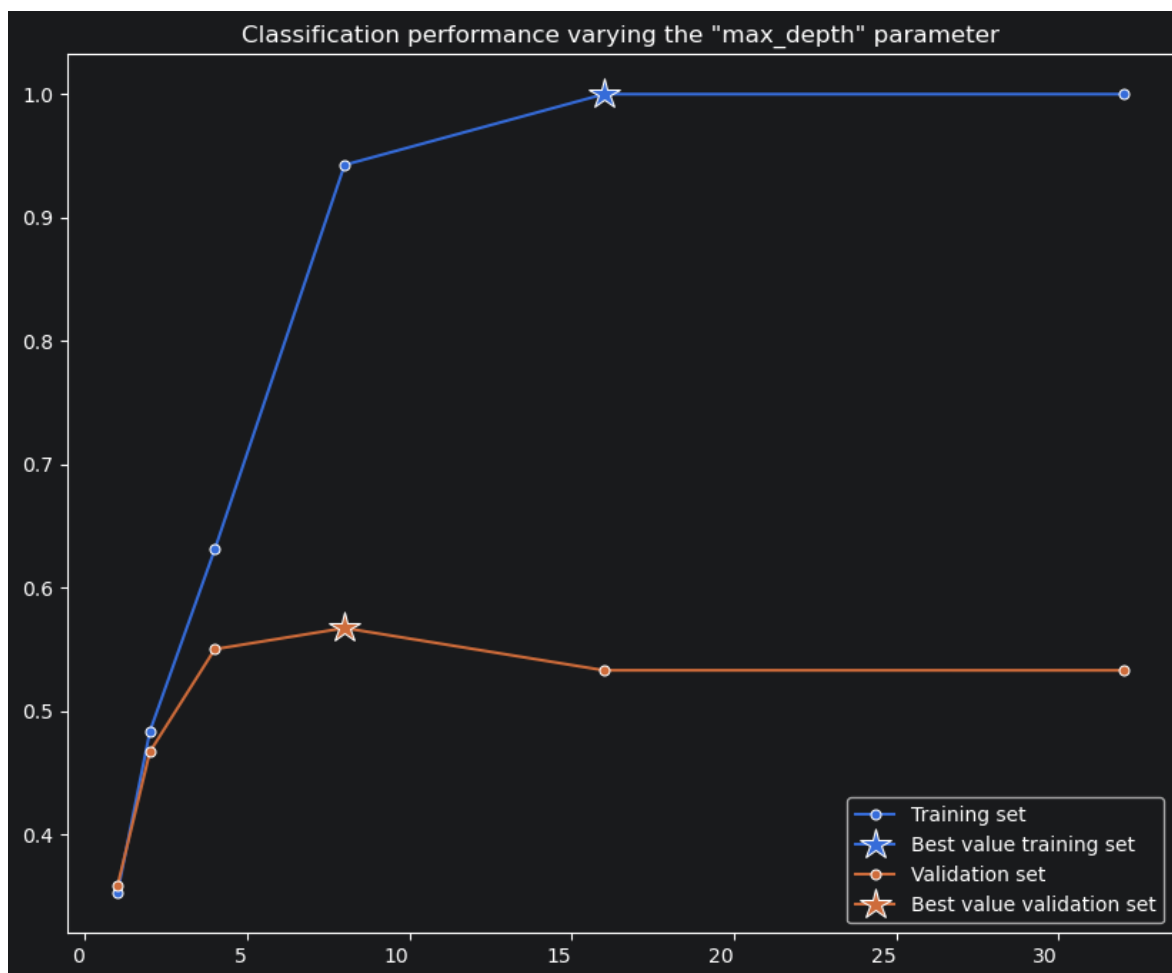
    # Drawing the confusion matrix
    uc.draw(what="confusion", set="test", title=algorithm, show=True)

    # Dendrogram
    if algorithm == "decisiontree":
        uc.draw(
            what="tree",
            title="Dendrogram of " + algorithm,
            max_depth=2,
            figsize=(24, 16),
            show=True
        )
)
```



Warning: failed to launch javaldx - java may not function correctly
convert /home/ac87/PycharmProjects/DMML/reports/pdf/decisiontree.html as a
Writer/Web document -> /home/ac87/PycharmProjects/DMML/reports/pdf/decisio
ntree.pdf using filter : writer_web_pdf_Export
Overwriting: /home/ac87/PycharmProjects/DMML/reports/pdf/decisiontree.pdf





Warning: failed to launch javaldx - java may not function correctly
convert /home/ac87/PycharmProjects/DMML/reports/pdf/randomforest.html as a Writer/Web document -> /home/ac87/PycharmProjects/DMML/reports/pdf/randomforest.pdf using filter : writer_web_pdf_Export
Overwriting: /home/ac87/PycharmProjects/DMML/reports/pdf/randomforest.pdf

Warning: failed to launch javaldx - java may not function correctly
convert /home/ac87/PycharmProjects/DMML/reports/pdf/svm.html as a Writer/Web document -> /home/ac87/PycharmProjects/DMML/reports/pdf/svm.pdf using filter : writer_web_pdf_Export
Overwriting: /home/ac87/PycharmProjects/DMML/reports/pdf/svm.pdf

